PCBC H Parametric & Sequences Test (Practice D) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A baseball is hit from a height of 3 feet at an angle of 50° with an initial speed of 90 ft/s.

a) Write a set of parametric equations for the motion of the baseball. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height of the baseball. At that time, how far has it traveled horizontally?

c) Determine how long the baseball is in the air and how far it travels.

d) Would the ball clear a 35-ft-high wall that is 220 ft from the batter?

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $2,\ \frac{11}{3},\ \frac{16}{3},\ 7,\ \frac{26}{3},\ \ldots$

Express the sum using summation notation. (3 points)

9. $\frac{1}{5}+\frac{1}{25}+\frac{1}{125}+\cdots+\frac{1}{15625}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{a=1}^{30}(20-a)$

11. $\sum_{k=1}^{20}\left(k^3-5\right)$

12. $\sum_{n=4}^{18} 3n^2$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x+y)^5$

14. $(2y-3)^5$

Extra practice: the other things Friday could ask

Not part of the 27 points. Worked in the key.

Projectile, one more part.

E1. Write the path of the baseball in #7 as a rectangular equation (eliminate t).

A sequence is defined recursively. Write the first five terms.

E2. $a_1 = -1; \quad a_n = n\,a_{n-1}$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E3. $1,\ -\frac{1}{9},\ \frac{1}{81},\ -\frac{1}{729},\ \ldots$

Write out the sum.

E4. $\displaystyle\sum_{n=1}^{4} \frac{n^2+1}{n}$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E5. $-2,\ 2,\ -2,\ 2,\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Baseball: $h = 3$, $\theta = 50°$, $v = 90$ ft/s. Degree mode, Graph in Parametric.

a) $x = (90\cos 50°)t$, $y = (90\sin 50°)t - 16t^2 + 3$ — *plug into the given formula; degree symbol*

$0 \le t \le 5$, tstep 0.01; x : 0 to 270, y : −5 to 85 — *window*

b) $\boxed{y \approx 77.270 \text{ ft}}$, $\boxed{x \approx 124.640 \text{ ft}}$ — *trace to the top ($t \approx 2.154$); read y for the height, x for how far*

c) $\boxed{t \approx 4.352 \text{ s}}$, $\boxed{x \approx 251.772 \text{ ft}}$ — *trace to $y \approx 0$*

$t = \frac{220}{90\cos 50°} \approx 3.803$ — *d) when is $x = 220$? solve x for t*

$y \approx (90\sin 50°)(3.803) - 16(3.803)^2 + 3 \approx 33.795$ — *height at that moment (or trace to $x \approx 220$)*

d) $\boxed{\text{No: } y \approx 33.795 \text{ ft} < 35 \text{ ft}}$ — *compare with the wall's height*

Units: feet and seconds.

8. $\overset{1}{2}, \overset{2}{\frac{11}{3}}, \overset{3}{\frac{16}{3}}, \overset{4}{7}, \overset{5}{\frac{26}{3}}, \ldots$

top: $5n + 1$, bottom: 3 — *write 1 2 3 4 above; top and bottom separately*

$\boxed{a_n = \frac{5n+1}{3}}$ — *rewrite everything over 3: $\frac{6}{3}, \frac{11}{3}, \frac{16}{3}, \frac{21}{3}, \frac{26}{3}$; tops go up by 5*

9. $\frac{1}{5} + \frac{1}{25} + \cdots + \frac{1}{15625}$

$\frac{1}{5^1}, \frac{1}{5^2}, \ldots, \frac{1}{5^6}$ — $15625 = 5^6$

$\boxed{\sum_{n=1}^{6} \left(\frac{1}{5}\right)^n}$ — *any letter works if it matches; $\sum \frac{1}{5^n}$ is the same*

10.

$\sum_{a=1}^{30} 20 - \sum_{a=1}^{30} a$ — *(3) split; the letter a works like k*

$20(30) - \frac{30(31)}{2}$ — *(5), (6) with $n = 30$*

$\boxed{135}$

11.

$\sum_{k=1}^{20} k^3 - 5\sum_{k=1}^{20} 1$ — *(2), (3) split; (1) constants out*

$\left[\left[\frac{20(21)}{2}\right]^2\right] - 5(20)$ — *(8), (5) with $n = 20$; then the calculator*

$\boxed{44000}$

12.

$3\sum_{n=4}^{18} n^2$ *(1)*

$3\left[\sum_{n=1}^{18} n^2 - \sum_{n=1}^{3} n^2\right]$ *(4): starts at 4, so $j=3$*

$3\left[\frac{18(19)(37)}{6} - \frac{3(4)(7)}{6}\right]$ *(7)*

$\boxed{6285}$

13. $(x+y)^5$: $n=5$, row 1 5 10 10 5 1. Two variables: same blanks.

$1x^5y^0 + 5x^4y^1 + 10x^3y^2 + 10x^2y^3 + 5x^1y^4 + 1x^0y^5$ *x powers count down, y powers count up*

$\boxed{x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5}$

Every term's powers add to 5.

14. $(2y-3)^5$: $n=5$, row 1 5 10 10 5 1. The sign goes with its term.

$1(2y)^5(-3)^0 + 5(2y)^4(-3)^1 + 10(2y)^3(-3)^2 + 10(2y)^2(-3)^3$ *blanks first: first term's powers count down, second term's count up*

$+5(2y)^1(-3)^4 + 1(2y)^0(-3)^5$

$(2y)^5 = 32y^5, \ \ldots$ *the whole first term gets the power*

$\boxed{32y^5 - 240y^4 + 720y^3 - 1080y^2 + 810y - 243}$

Second term negative: odd powers of -3 are negative, so signs alternate.

Extra practice (not scored)

E1. path of #7 as a rectangular equation

$t = \frac{x}{90\cos 50^\circ}$ *solve x for t*

$y = (90\sin 50^\circ)\frac{x}{90\cos 50^\circ} - 16\left(\frac{x}{90\cos 50^\circ}\right)^2 + 3$ *substitute*

$\boxed{y = -\frac{16x^2}{8100\cos^2 50^\circ} + (\tan 50^\circ)x + 3}$ $\frac{\sin}{\cos} = \tan$; $\approx -0.00478x^2 + 1.192x + 3$

E2. $a_1 = -1$; $a_n = n\,a_{n-1}$

$a_2 = 2(-1) = -2, \ \ a_3 = 3(-2) = -6$

$a_4 = 4(-6) = -24, \ \ a_5 = 5(-24) = -120$

$\boxed{-1, \ -2, \ -6, \ -24, \ -120}$

E3. $\overset{1}{1}, \ \overset{2}{-\frac{1}{9}}, \ \overset{3}{\frac{1}{81}}, \ \overset{4}{-\frac{1}{729}}, \ldots$

top: 1, bottom: 9^{n-1} — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = \left(-\frac{1}{9}\right)^{n-1}}$ — *each term is $-\frac{1}{9}$ times the last; starts at power 0*

E4. $\sum_{n=1}^{4} \frac{n^2+1}{n}$

$\frac{1+1}{1} + \frac{4+1}{2} + \frac{9+1}{3} + \frac{16+1}{4}$ — *upper index is 4: write all four terms, no ellipsis*

$\boxed{2 + \frac{5}{2} + \frac{10}{3} + \frac{17}{4}}$

E5. $\overset{1}{-2},\ \overset{2}{2},\ \overset{3}{-2},\ \overset{4}{2}, \ldots$

top: 2, bottom: 1 — *write 1 2 3 4 above; top and bottom separately*

$-,+ \Rightarrow (-1)^n$ — *first term negative*

$\boxed{a_n = 2(-1)^n}$